User guide

How to run Wellbeing Lab, what every page shows, and how to get predictions out of it, with screenshots of the real app.

Wellbeing Lab · Author: Vincent Mui

Quick start

Wellbeing Lab asks one question: **is the time people spend on social media related to their mental wellbeing?** It cleans a survey of 1,500 people, compares six classification models, and shows the answer as a dashboard you can also query as an API.

1. From the project folder run `./start.sh`. The first run trains the models (about a minute) and installs the front end.
2. Open `http://localhost:5180` in a browser. The green **API online** badge at the top right means the page can reach the server.
3. Start on Overview for the headline result, read the step-by-step analysis on Notebooks, then try Predict.
4. When you are done run `./stop.sh` (or press Ctrl-C in the terminal that runs `start.sh`).

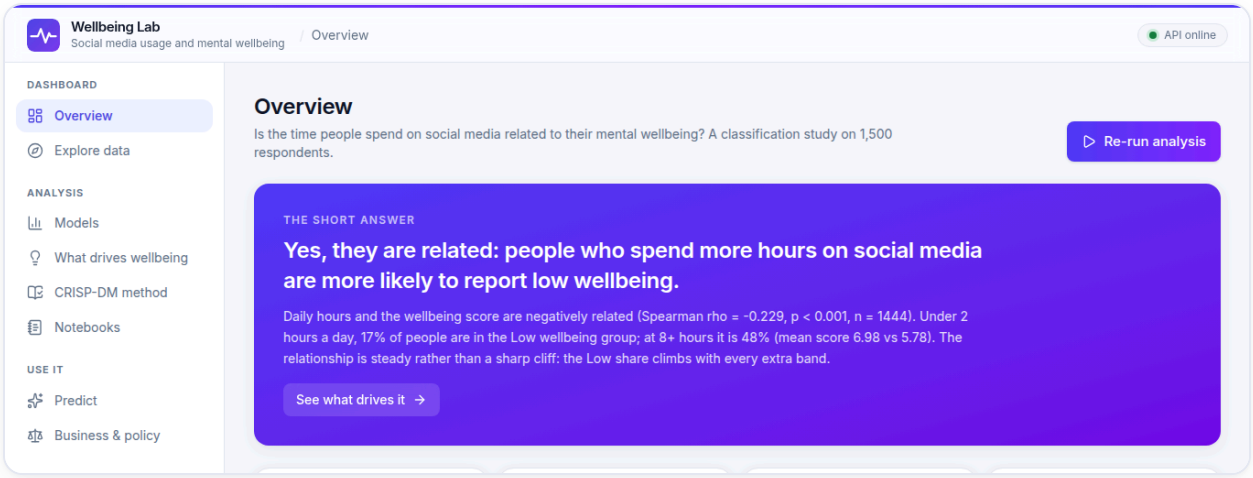
> The sidebar groups the pages by purpose. The **Collapse** button at its bottom shrinks it to icons, and the browser remembers your choice. On a phone the menu button at the top left opens it.

Tour of the pages

The pages follow the CRISP-DM order: understand the data, build models, interpret them, then use them. Click any screenshot to see it full size.

Overview

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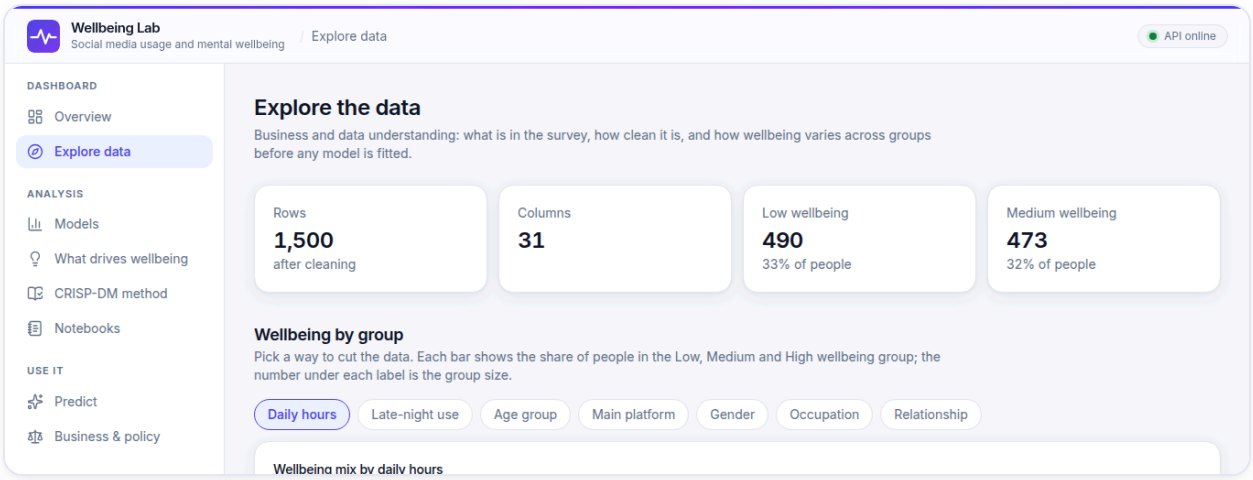
Overview: the answer in one screen.

- The tiles give the number of respondents, how strongly hours of use and wellbeing move together (Spearman ρ), the best model, and its accuracy on people it never saw.
- The chart shows how the share of Low, Medium and High wellbeing changes as daily hours rise. The small chart beside it shows how many people each bar is based on, so you can tell a firm result from a thin one.

- Key findings are written from the measured numbers each time the analysis runs, so they always match the charts.
- Re-run analysis (top right) retrains everything. See Re-running the analysis below.

Explore data

Open page ↗



Explore data: what is in the survey before any model is fitted.

- The top tiles show the rows and columns left after cleaning and the size of the Low and Medium groups. Data-quality detail (blank cells per column) is at the bottom.
- Wellbeing by group: choose how to cut the data (hours of use, age, late-night use, platform, gender, occupation, relationship status). Each bar shows the share of Low, Medium and High, and the number under each label is how many people are in that group.
- The hours-and-late-night chart compares average wellbeing with and without late-night use. The gap between the lines is the late-night effect.
- The correlation chart ranks measures by how closely they move with the wellbeing score. Grey bars are the mental-state measures such as anxiety and sleep quality. They are outcomes rather than habits, so the models do not use them as inputs.

Models

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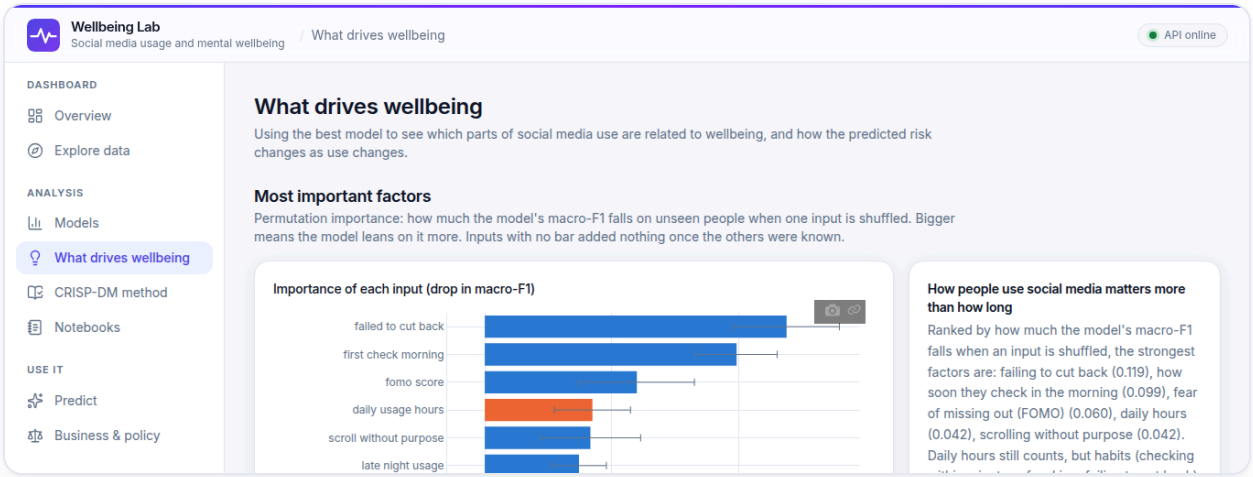
Models: six classifiers tuned by grid search and compared on people they never saw.

- The highlighted card names the best model and its best hyper-parameters. It is chosen on cross-validated macro-F1 from the training data only, so the test set plays no part in the choice.

- The table lists every model with its tuned hyper-parameters, cross-validated score and test metrics. The always-guess-the-biggest-group baseline is the bar to beat.
- The bar chart sets cross-validated against test macro-F1. Two bars close together mean the model generalises. A big gap (the decision tree and forest here) means it overfits.
- Decision-threshold tuning shows the per-class weights that were searched, the before and after scores, and whether they were adopted. Use the Standard rule / Tuned weights switch to see the confusion matrix and per-group results for each.

What drives wellbeing

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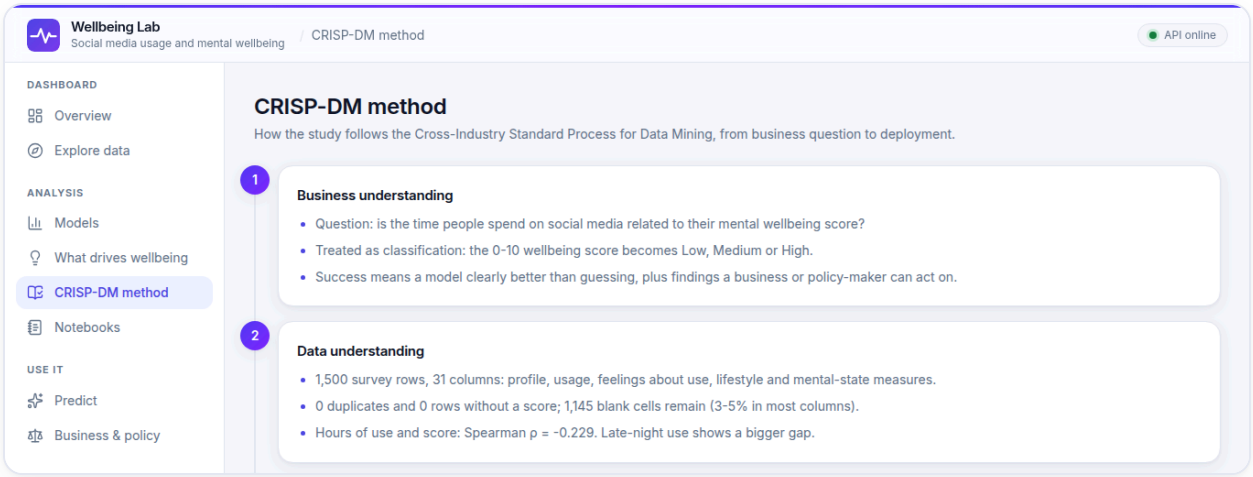


What drives wellbeing: which habits the best model leans on, and how predicted risk moves.

- Importance shows how much the model's score drops when one input is shuffled. A long bar means the model relies on that input.
- The risk curves re-score every test person as if they used social media for 1, 2, 3 … hours, keeping everything else about them fixed. The same is done for late-night use and for notifications.
- These are associations in one survey. They do not prove that social media causes lower wellbeing.

CRISP-DM method

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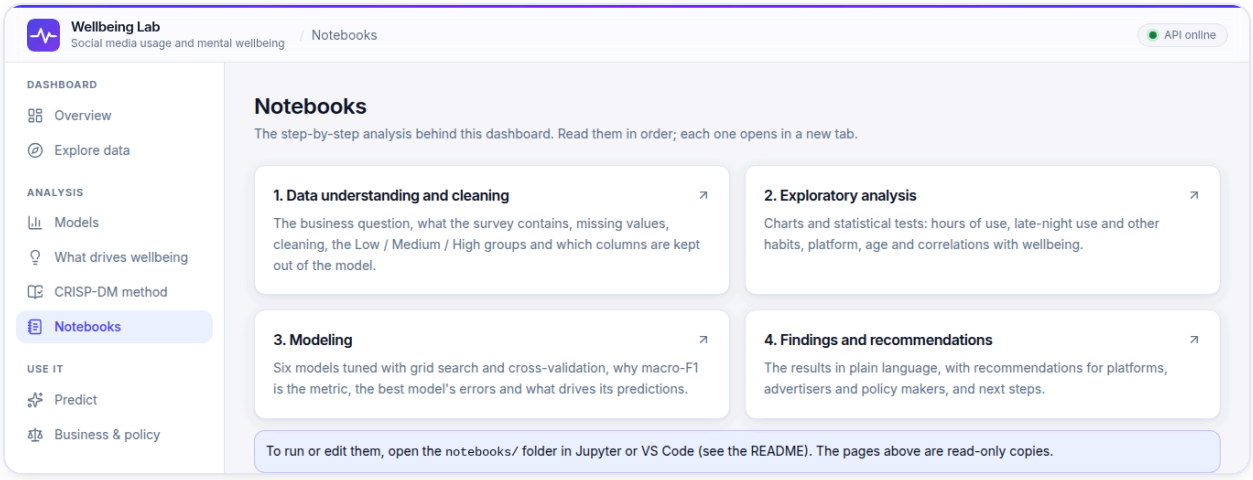


CRISP-DM method: the six stages, with the numbers from this run.

Business understanding, data understanding, preparation, modelling, evaluation and deployment, each with what was done and what it found. Use it as the written method for a report.

Notebooks

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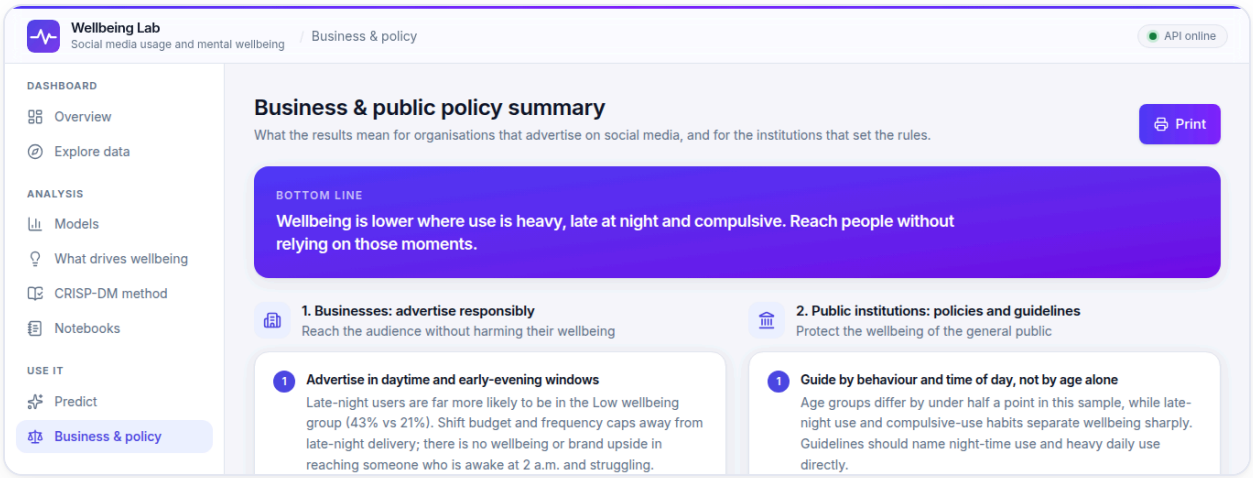


Notebooks: the four analysis notebooks, readable in the browser.

- Four cards, in reading order: data understanding and cleaning, exploratory analysis, modeling, and findings and recommendations. Each opens a read-only HTML copy in a new tab, with all charts and results included.
- The notebooks are the technical write-up of the study. They use the same code as the app, so their numbers match the dashboard.

Business & policy

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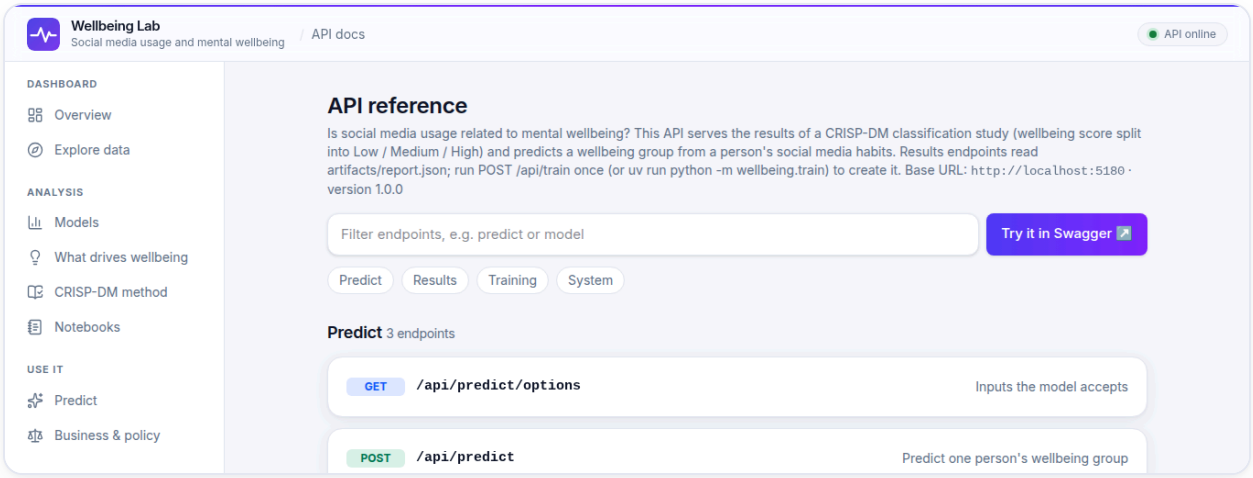


Business & policy: recommendations that rest on the measured results.

- Two sections: how businesses can advertise responsibly with audience wellbeing in mind, and how public institutions can set policy and guidelines.
- Each recommendation quotes the figure it is based on. A short limits section says what the data cannot show.
- Print (top right) opens the browser's print dialog, so you can print the page or save it as a PDF.

API docs

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API docs: every endpoint with its parameters, a ready-to-copy curl command and an example response.

The page is built from the server's own description, so it cannot fall out of date. The interactive Swagger version is at /api/docs.

The notebooks

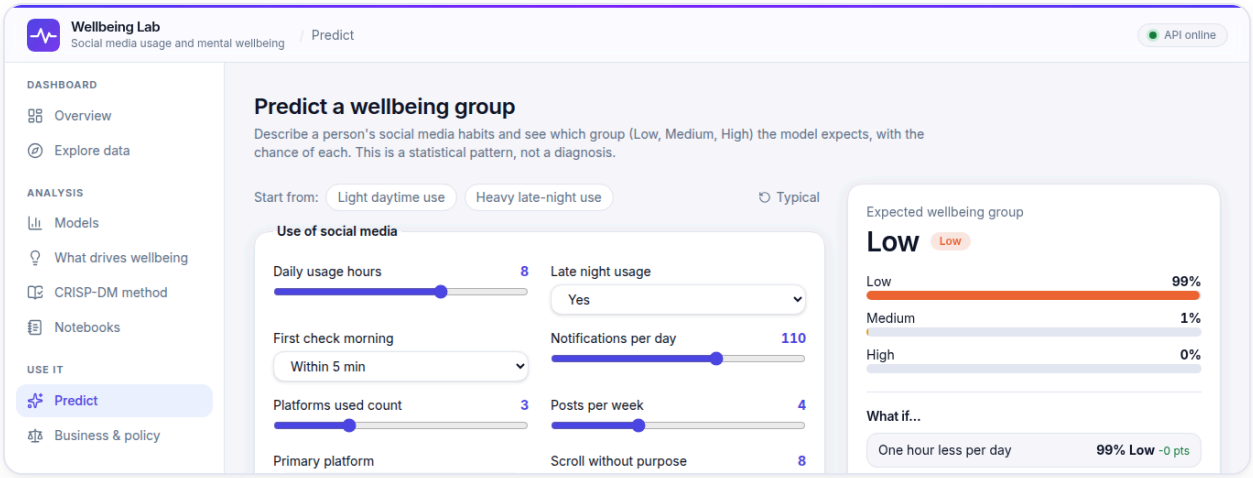
The analysis is also delivered as four Jupyter notebooks in the `notebooks/` folder. Read them in order:

NOTEBOOK	WHAT IT COVERS
01 Data understanding and cleaning	The business question, the survey columns, missing values, cleaning, the Low / Medium / High groups and which columns are excluded to avoid leakage.
02 Exploratory analysis	Charts and statistical tests (Spearman correlation, chi-square) for hours of use, habits, platform, age and the wellbeing score.
03 Modeling	Six models tuned by grid search and cross-validation, the macro-F1 metric and why it was chosen, the confusion matrix, permutation importance and the threshold-tuning check.
04 Findings and recommendations	Plain-language findings, recommendations for platforms, advertisers and policy makers, and next steps.

1. To read them without installing anything, open Notebooks in the app.
2. To run or edit them, run `uv sync` once, then `uv run jupyter lab notebooks/` (or open the folder in VS Code). Run each notebook from top to bottom; the full set takes under a minute.
3. After changing a notebook, run `scripts/notebooks-html.sh` to refresh the copies shown in the app.

Making predictions

One person at a time

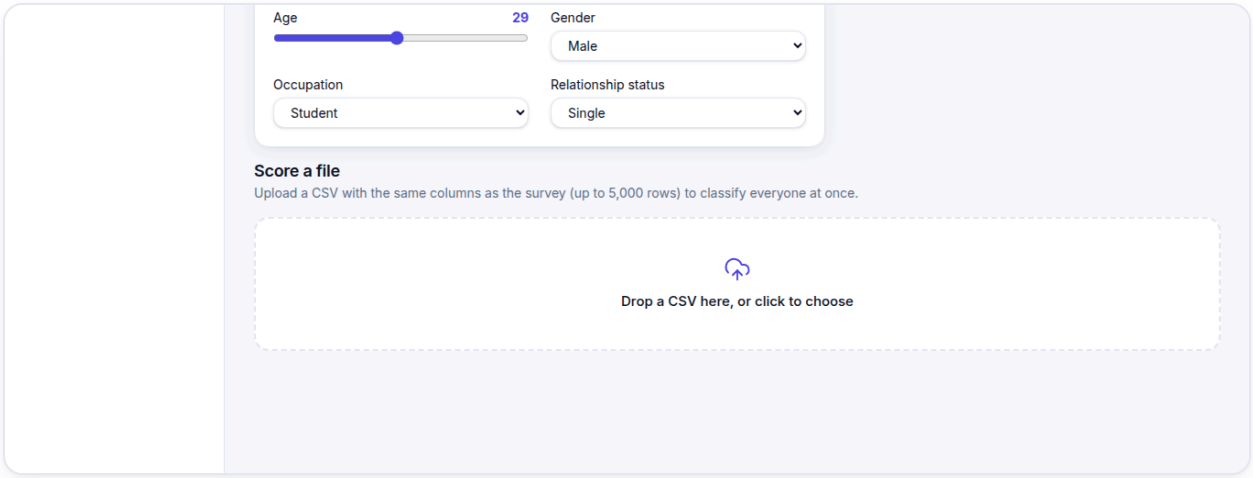


Predict, started from the Heavy late-night use preset. The sliders are on the left, the result and what-if scenarios on the right.

1. Open Predict. The form starts from a typical respondent (the most common answer or the median for each field).
2. Optionally pick a preset under **Start from**: *Light daytime use* or *Heavy late-night use*. **Typical** resets everything.
3. Move the sliders and choose from the lists. The result on the right updates about 0.2 seconds after you stop, so there is no submit button.
4. Read the **Expected wellbeing group** and the three probability bars. They add up to 100%.
5. Check **What if...**: it re-scores the same person with one hour less per day, and with no late-night use (shown only when it applies). The figure is the chance of Low wellbeing and the change in percentage points. Green means the change helps.

- Leave a list on *Not specified* if you do not know the answer. The model fills the gap the same way it did in training.
- The result is a statistical pattern in a survey. It is not a diagnosis of any person.
- A link such as `/predict?preset=Heavy late-night use` opens the page with that preset already applied.

A whole file at once



Score a file: drop a CSV on the dashed area, or click it to choose one.

1. Prepare a CSV with a header row. Column names should match the survey's (for example `Daily_Usage_Hours`, `Late_Night_Usage`). Capitalisation, spaces versus underscores, and column order do not matter.
2. Drop the file on **Score a file** at the bottom of the Predict page, or click the area to choose it.
3. The page shows how many people were scored and how many fell in each group. **Download results** saves `predictions.csv` with the row number, predicted group and the three probabilities.

Download a 25-row sample CSV

RULE	DETAIL
Size	Up to 5,000 rows per file.
Columns	Any survey column can be used. Extra columns such as User_ID, Mental_Wellbeing_Score and Addiction_Level are ignored, and missing columns are treated as blank.
Blank cells	Allowed. They are filled the same way as in training.
No match	If none of your column names is recognised the upload is rejected with the list of expected names.

Re-running the analysis

Run the analysis again after replacing the data file, changing a setting, or if you simply want to see it work. It cleans the data, tunes all six models, picks the best, tunes the threshold, and rewrites the report and model. Every page, including the written findings, then updates.

1. Click **Re-run analysis** at the top right of Overview or Models. On a fresh install with no results, the same button appears as **Run the analysis** in the middle of the page.
2. Watch the progress bar and step name. It takes under a minute.
3. The pages refresh with the new numbers when it finishes. If it fails, the message appears under the button.

Only one run can be active at a time. A second request gets a "run already in progress" answer. From a terminal use:

```
uv run python -m wellbeing.train            # full run
uv run python -m wellbeing.train --quick    # small grids, for a fast check
```

Using the API

The UI is a client of the same API, so everything on screen can be scripted. Interactive docs are at /api/docs, and the machine-readable description is at /api/openapi.json.

Predict one person

Send only the fields you know, using the names listed by `GET /api/predict/options`. Unknown names, unknown category values and non-numbers are rejected with a 422 that says what is allowed.

```
curl -X POST http://localhost:5180/api/predict \
  -H "Content-Type: application/json" \
  -d '{"inputs": {"daily_usage_hours": 6.5, "late_night_usage": "Yes",
                  "first_check_morning": "Within 5 min", "failed_to_cut_back": "Yes",
                  "fomo_score": 7}}'
```

```
{
  "prediction": "Low",
  "probabilities": { "Low": 0.9377, "Medium": 0.0617, "High": 0.0007 },
  "what_if": [
    { "scenario": "One hour less per day", "prediction": "Low", "probabilities": { "Low": 0.9207, "Medi
    { "scenario": "No late-night use",     "prediction": "Low", "probabilities": { "Low": 0.72, "Mediur
  ]
}
```

Predict a file

```
curl -X POST http://localhost:5180/api/predict/batch -F "file=@people.csv"
```

The reply has `rows`, `counts` per group, and `results` with a prediction and probabilities for each row in the file's order. Errors: 400 if the file is not a CSV, 413 above 5,000 rows, 422 if no column is recognised.

Re-train

```
curl -X POST http://localhost:5180/api/train          # -> {"job_id": "3f2c9a1e", "status": "running"}
curl http://localhost:5180/api/jobs/3f2c9a1e        # poll until "status" is "done" or "failed"
```

ENDPOINT	RETURNS
GET /api/health	Liveness check used by the API online badge.
GET /api/report	The whole report in one document.
GET /api/models, /threshold, /insights	Model comparison, threshold tuning, importance and risk curves.
GET /api/eda, /data/summary	Exploratory statistics and data-quality numbers.
GET /api/business-summary	Recommendations with the numbers they rest on.
Before the first run	Result endpoints answer 404 with a hint to run the analysis.

Settings

Everything works with no configuration. To change something copy `.env.example` to `.env` and uncomment the line. Restart after editing, then re-run the analysis if the setting affects the model.

SETTING	DEFAULT	MEANING
DATA_FILE	data/social_media_addiction_mental_wellbeing.csv	The survey CSV to analyse.
ARTIFACTS_DIR	artifacts	Where the trained model and report are saved.
APP_NAME	Wellbeing Lab	Name shown in the header and page title.
TEST_SIZE	0.25	Share of people held out to score the final model.
CV_FOLDS	5	Folds used by the grid searches.
N_FEATURES	20	How many of the most informative columns each model keeps.
API_PORT / UI_PORT	8010 / 5180	Ports used by start.sh and stop.sh (set them in the shell, for example UI_PORT=5200 ./start.sh).

Reading the numbers

TERM	WHAT IT MEANS HERE
Low / Medium / High	The wellbeing score split into thirds: the lowest third is Low, the highest third High. The median score sits inside Medium.
Macro-F1	The F1 score of Low, Medium and High averaged equally, so a model cannot score well by ignoring one group. 1 is perfect. Always guessing the biggest group scores about 0.18.
Accuracy	The share of people placed in the right group.
ROC AUC	How well the predicted probabilities rank people, from 0.5 (chance) to 1 (perfect).
Cross-validated (CV)	Scores from splitting the training data five ways, training on four parts and testing on the fifth. It is used to choose between models.
Test	Scores on the held-out people, used once at the end. This is the honest estimate.
Hyper-parameter	A model setting chosen before training, such as C for logistic regression. Grid search tries many and keeps the best.
Decision threshold	With three groups, a weight on each group's probability before the largest is picked. Tuned weights are adopted only if they help.
Permutation importance	How far macro-F1 falls when one input is shuffled. A bigger fall means a more important input.
Association	Two things moving together. It does not show that one causes the other.

Troubleshooting

SYMPTOM	WHAT TO DO
Badge says API offline	The server is not running or not reachable. Run ./start.sh, and check the terminal for errors.
Pages show “No results yet”	No analysis has run. Click Run the analysis, or run uv run python -m wellbeing.train.
A port is already in use	Another program has 8010 or 5180. Start with different ones: API_PORT=8020 UI_PORT=5200 ./start.sh.
Re-run says a run is in progress	Wait for the current one to finish (under a minute) and try again.
Upload rejected with 422	None of the column names was recognised. Compare your header with the sample CSV.
Upload rejected with 413	The file has more than 5,000 rows. Split it.
Prediction field refused	The message lists the allowed values. Category names must match exactly, for example “Within 5 min”.
Stop everything	Run ./stop.sh. It only stops this project's own processes.